

Task 3 - Technical Appendix

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Data generation

The prototype generates 5,000 synthetic credit applications with applicant features and a synthetic default outcome. It then trains the primary logistic-regression model and uses the trained model's `predict_proba` output as `model_score`. Approval decisions, manual-review flags and ranked decision reasons are derived after scoring. The data is intentionally synthetic because real cross-border credit data with protected attributes is commercially sensitive and usually unavailable for coursework.

Risk logic

Default risk increases when credit score is lower, debt-to-income is higher, income is lower, prior default is present, regional portfolio risk is higher and employment is weaker. A mild controlled protected-group score shift is included so the tool can demonstrate fairness monitoring without inventing an extreme or unrealistic disparity.

Model implementation

The generation script trains a logistic-regression primary model and a Random Forest challenger. The application-level `model_score` column is the trained logistic model's predicted default probability, not a synthetic pre-model score. Logistic performance: AUC 0.800, accuracy 0.876. Random Forest performance: AUC 0.793, accuracy 0.785. Logistic regression remains the primary governance model because its feature contributions can be mapped cleanly to adverse-action reasons.

Task 3 - Metric Formulas

Approval parity ratio

$\text{approval_parity_ratio} = \text{approval_rate}(\text{protected group}) / \text{approval_rate}(\text{reference group})$.

In the generated portfolio, protected-group approval is 0.7305, reference-group approval is 0.8250 and approval parity is 0.8855.

False positive rate by group

In this credit context, a false positive means the tool rejects or flags a low-risk applicant who did not default in the synthetic outcome. The group-level false positive rate checks whether one group bears more erroneous denials or review burdens.

False negative rate by group

A false negative means the model approves an applicant who later defaults in the synthetic outcome. This metric matters because a tool that only optimizes consumer approvals may ignore bank safety and soundness risk.

FNR tension in the generated data

The protected-group FNR is 0.3937, while the reference-group FNR is 0.4706. This means the protected group has fewer actual defaulters incorrectly approved, but also faces a lower approval rate overall. The governance question is therefore not simply whether one group is more risky; it is whether the model has become more conservative for that group in a way that shifts consumer burden.

Task 3 - Jurisdiction Logic

US output

US mode focuses on adverse-action reason quality and internal fair-lending risk evidence. The tool is careful not to say that a statistical disparity is automatically a legal violation, especially because the 2026 CFPB Regulation B final rule changes the treatment of disparate impact. This is why the US output says monitoring or review rather than legal conclusion.

EU output

EU mode treats AI creditworthiness evaluation as high-risk AI governance. The tool therefore emphasizes bias testing, documentation, human oversight and remediation ownership. This output is not just the US report translated into EU language; it is a different governance package.

EU AI Act Article 10, 14 and 17 mapping

Article 10 is implemented as data-governance evidence: data suitability, feature definitions, jurisdiction fields, protected-group monitoring, bias metrics and sensitivity checks. Article 14 is implemented as reviewer sign-off and human-oversight evidence. Article 17 is implemented as quality-management evidence: rule versions, effective dates, validation results, conflict notes, limitation records and monitoring cadence.

Contradictory rules

If two jurisdictions impose different expectations, the prototype does not average them. It uses the active jurisdiction for the formal output and records a conflict note so a human governance owner can decide whether the stricter evidence trail should be retained for cross-border re-use.

Task 3 - Sensitivity Result

Sensitivity result

The sensitivity test shifts protected-group model scores and recalculates parity under US and EU thresholds. Selected scenarios:

Shift 0.00: parity 0.8855; US flag False; EU flag True.

Shift 0.02: parity 0.8311; US flag False; EU flag True.

Shift 0.04: parity 0.7975; US flag True; EU flag True.

Task 3 - Data Dictionary Extract

The full field-level dictionary is exported as data/Task3_data_dictionary.csv. This page highlights the production-risk fields a reviewer should challenge first.

Field	Model input?	Protected/proxy?	Production stance
protected_group	False	Protected/proxy	Fairness analysis only; do not use as a production credit model input.
credit_score	True	Potential proxy	Potentially valid only with bureau lineage, permissible-purpose checks and adverse-action mapping.
region_risk_score	True	Potential proxy	High-risk proxy; avoid or strictly validate before production scoring.
model_score	False	Derived	Model output; audit and monitor, not an independent input.

Audit note Protected attributes and proxies are used for monitoring and explanation governance, not as production credit-model inputs unless separately approved.

Task 3 - Limits and Production Hardening

What the prototype does not prove

It does not prove Citi's actual compliance posture, does not use real customer data, does not provide legal advice, does not decide credit applications autonomously and does not automatically repair model bias. These limits are not excuses; they are part of the governance design.

Production hardening needed

A real deployment would need validated source systems, model inventory integration, access control, audit logs, legal-change workflow, adverse-action notice review, drift monitoring, appeal workflow and documented accountability for overrides.

Why the limitation is acceptable for this assignment

The assignment asks for a working prototype or mock-up that demonstrates jurisdiction-aware logic, not a production bank platform. The submitted tool therefore prioritizes transparent regulatory mechanics, reproducible synthetic data and honest boundaries over unsupported claims of deployment readiness.